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UTS MTK SAINS DATA

Parameter Personalisasi

1. menentukan N
   $N = 31$
2 menentukan a
   $a = \text{Flor}\left(\frac{31}{10}\right) = 3$
3 menentukan b
   $b = 31 \bmod 10 = 1$

4 hitung k
   $k = a + b + 1$
   $k = 3 + 1 + 1 = 5$

5 hitung θ
   $\theta_0 = (a \times 30) + (b \times 5)$
   $\theta_0 = (3 \times 30) + (1 \times 5) = 90 + 5 = 95^\circ$

6 hitung α
   $\alpha = 0.001 \times (b + 1)$
   $\alpha = 0.001 \times (1 + 1) = 0,002$

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$k = 5,\ a = 3,\ b = 1$

5.1

ⓐ. Hitung

L_1

$$\lim_{x \to k} \frac{x^2 - k^2}{x - k}$$

Faktor

$$= \lim_{x \to k} \frac{(x-k)(x+k)}{x-k} = \lim_{x \to k} (x+k) = 2k = 10$$

L_2

$$\lim_{x \to 0} \frac{\sqrt{x + (k+1)^2} - (k+1)}{x}$$

rasionalisasi $= \dfrac{x}{x\left(\sqrt{x+(k+1)^2} + (k+1)\right)}$

$$= \frac{1}{\sqrt{x + (k+1)^2} + (k+1)}$$

$$\to \frac{1}{2(k+1)} = \frac{1}{12}$$

L_3

$$\lim_{x \to \infty} \frac{(a+2)x^3 + bx + 7}{x^3 - x + k}$$, ambil suku tertinggi $= \dfrac{a+2}{1} = 5$

5.2 Limit trigonometri

ⓒ

L_4

$$\lim_{x \to 0} \frac{\sin((a+1)x)\tan((b+2)x)}{x^2}$$

gunakan pendekatan
$\sin(kx) \sim kx$
$\tan(kx) \sim kx$

$$= \frac{(a+1)x\,(b+2)x}{x^2} = (a+1)(b+2) = 4 \cdot 3 = 12$$

Berdasarkan limit dasar

$$\lim_{x \to 0} \frac{\sin x}{x} = 1, \quad \frac{\tan x}{x} = 1$$

5.3 Limit tak hingga

ⓓ

$$R(n) = \frac{100n + k}{n + (a+b+1)}$$ pendapatan 100jt sebulan

$$\lim_{n \to \infty} = \frac{100}{1} = 100$$

ⓔ $= \dfrac{(k-b)n}{\sqrt{n^2 + kn} + \sqrt{n^2 + bn}}$

selisihnya $\to$ konvergen dan konstanta tetap(2)

$$= \frac{(k-b)n}{n\left(\sqrt{1 + k/n} + \sqrt{1 + b/n}\right)}$$

$$\to \frac{k-b}{2} = \frac{5-1}{2} = 2$$

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5.4. Kekontinuan

Fungsi

$$C(x) \begin{cases} \dfrac{x^2-k^2}{x-k} & x < k \\ p, & x = k \\ ax+q, & x > k \end{cases}$$

(f) ~~Kon~~ syarat kontinu

1. Limit kiri = nilai fungsi
2. Limit kanan = nilai fungsi
3. limit kiri = limit kanan

Limit kiri :

$$\lim_{x \to k^-} = x + k = 2k$$

Limit kanan :

$$\lim_{x \to k^+} = ak + q$$

Samakan

$$p = 2k$$

$$ak + q = 2k \rightarrow q = 2k - ak = k(2-a)$$

hasil

- $p = 2k = 10$
- $q = k(2-a) = 5(2-3) = -5$

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